

EXPERIMENT 22 : DATA SEGMENTATION BY K-MEANS CLUSTER USING WEKA AND R-TOOL

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to the Iris dataset. The 'Clusterer' tab is selected, and the 'SimpleKMeans' algorithm is chosen. The 'Cluster mode' is set to 'Use training set'. The 'Result list' on the left shows the results for the 190436 - SimpleKMeans model. The 'Clusterer output' pane displays the following information:

Cluster mode
☒ Use training set
☐ Supplied test set
☐ Percentage split % 66
☐ Classes to clusters evaluation (Nom) class
☒ Store clusters for visualization

Clusterer output

Number of iterations: 7
Within cluster sum of squared errors: 62.1436882815797

Initial starting points (random):
Cluster 0: 6.1,2.9,4.7,1.4,Iris-versicolor
Cluster 1: 6.2,2.9,4.3,1.3,Iris-versicolor

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (150.0)	Cluster# 0 (100.0)	Cluster# 1 (50.0)
sepalwidth	5.8433	6.262	5.006
sepalwidth	3.054	2.872	3.418
petalwidth	3.7587	4.906	1.464
petalwidth	1.1987	1.676	0.244
class	Iris-setosa Iris-versicolor	Iris-setosa	

Time taken to build model (full training data) : 0 seconds

==== Model and evaluation on training set ====

Clustered Instances

Cluster	Count	Percentage
0	100	(67%)
1	50	(33%)

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